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Anna Guseva

Education

26/02/2018 **Ph.D.**, University of Bremen, Germany
Thesis: "Dynamics and transport of instabilities in magnetized quasi–Keplerian Taylor–Couette flows", supervised by Prof. Marc Avila and funded by German Research Foundation

19/06/2014 **M. Sc. Thermophysics & molecular physics**, National Research University "Moscow Power Engineering Institute", Russia
Thesis: "Nonlinear analysis of the azimuthal magnetorotational instability"

19/06/2012 **B. Sc. Thermophysics**, National Research University "Moscow Power Engineering Institute", Russia, *Thesis*: "Electrovortex flow in hemispherical geometry"

Employment

Since 09/2025 **Assistant Professor**, Department of Fluid Mechanics, Polytechnic University of Catalonia (UPC), Spain
Affiliated with the Catalan Institute of Space Sciences (IEEC)
Member of Spanish Astronomical Society (SEA)

01/07/2023 – 31/08/2025 **Postdoctoral Fellow**, Laboratory for Astrophysics Research and Instrumentation (LIRA) and Laboratoire d'Etudes du Rayonnement et de la Matière en Astrophysique et Atmosphères (LERMA), the Paris Observatory, France
- Modelling of stellar magnetohydrodynamics as a part of ANR project PROMETHEE on star formation

01/09/2020 – 30/06/2023 **Marie Curie Research Fellow**, University of Leeds, UK
School of Mathematics
- Data-driven analysis and reduced-order modelling of astrophysical dynamos, in collaboration with Prof. S. Tobias

01/02/2018– 31/07/2020 **Postdoctoral researcher**, Polytechnic University of Madrid, Spain
School of Aeronautics and Space Engineering
- A comprehensive study of interaction between large- and small-scale structures in turbulent flows, in collaboration with Prof. J. Jiménez

01/03/2016 – 31/12/2017 **Research Assistant**, Center of Applied Space Technology and Microgravity, University of Bremen, Germany
- I found the possibility of a fully nonlinear self-sustained dynamo action in quasi-Keplerian accretion disc shear flows

01/08/2014 – 29/02/2016 **Research Assistant**, Institute of Fluid Mechanics, Friedrich-Alexander-University Erlangen-Nürnberg, Germany
- A study of transition to turbulence and momentum transport in magnetized quasi-Keplerian accretion disc flows

Awards

2025 A-scored ERC Starting Grant proposal
I was interviewed for an ERC Starting Grant with a project "Data-driven modelling of stellar shear flows" (PE9 - Universe Sciences), which obtained a final panel score A: fully meets the ERC's excellence criterion. Not funded for budgetary restrictions.

2024 & 2025 Severo Ochoa Visiting Grants
I was awarded funding to develop a collaboration on the magnetism of pulsating stars at the Institute of Astrophysics of Andalusia, Spain (IAA-CSIC) through Severo Ochoa Visitor Program.

2023 Postdoctoral Fellowship of the Paris Observatory
Peer-reviewed independent 2-years project on data-driven modelling of stellar magnetic activity for PLATO mission, with the aim of facilitating detection of exoplanets.

2022 The Observatoire de Paris – PSL Visiting Fellowship
I was awarded this fellowship to continue my research in the field of data-driven modelling of stellar and planetary dynamos, and participated in activities of the Graduate Program in Astrophysics at the Paris Observatory and PSL University.

2020 Marie Sklodowska Curie Individual Fellowship
I was awarded funding for implementation of the research project DynMode under Horizon 2020 Framework Programme. The projected was devoted to modelling of interscale nonlinear interactions in the dynamo flows using the novel data-driven approach.

2017 Kavli Institute for Theoretical Physics (US) Affiliate Visitor Grant
I was invited as an Affiliate Visitor to a program entitled "Recurrent Flows: The Clockwork Behind Turbulence", gathering experimentalists, theorists and computationalists to work on understanding the transition to turbulence.

2013 Professor Klaus Riedle-Foundation annual prize and scholarship
Professor Riedle Scholarship aimed to enhance knowledge exchange between Friedrich-Alexander-University Erlangen-Nürnberg and Moscow Power Engeneering Institute, and allowed me to perform a 6-months academic stay in Erlangen (Germany).

Preprints

2026 **A. Guseva**, C .Skene, and S. M. Tobias.
Data-driven discovery of dynamo cycle equations
arXiv preprint arXiv:2603.29079.
`https://arxiv.org/pdf/2603.29079`

Publications

19 peer-reviewed publications, 11 as a first author

2026 V. Durepaire, L. Petitdemange, K. Belkacem, **A. Guseva**, L. Manchon, R. Hollerbach and F. Daniel.
Magnetohydrodynamic instabilities in stellar radiative regions: I. Linear study of shear-driven instabilities
Astronomy & Astrophysics, 707, A167.
`https://doi.org/10.1051/0004-6361/202558545`

2025 **A. Guseva**, L .Manchon, L. Petitdemange and C. Pinçon.
Radial differential rotation leading to dipole collapse in pre-main-sequence stars
Astronomy & Astrophysics, 07, A45.
`https://doi.org/10.1051/0004-6361/202557616`

A. Guseva, L. Petitdemange and C. Pinçon.
Magnetic field morphologies in convective zones influenced by a turbulent surface layer
Astronomy & Astrophysics, 699, A68.
`https://doi.org/10.1051/0004-6361/202553685`

A. Guseva, L. Petitdemange and S. M. Tobias.
Run-away transition to turbulent strong-field dynamo
Journal of Geophysical Research: Planets, 130, e2024JE008496.
https://arxiv.org/abs/2405.10981

2024 **A. Guseva.**
Data-driven scale identification in oscillatory dynamos
Monthly Notices of the Royal Astronomical Society, vol. 528, no. 2 (2024).
https://doi.org/10.1093/mnras/stae079

C. Pinçon, L. Petitdemange, R. Raynaud, L. J. Garcia, **A. Guseva**, M. Rieutord, and E. Alecian.
Coriolis darkening in late-type stars. II. Effect of self-sustained magnetic fields in stratified convective envelopes
Astronomy and Astrophysics-A&A, 685, A129 (2024)
https://doi.org/10.1051/0004-6361/202349051

A. Guseva, B. Snow, and Z. Wang.
Data-driven modelling of coherent structures in mixing layers
Proceedings of the NFFDy Summer Programme on 'Data in Fluids' (2024).
https://doi.org/10.17863/CAM.107271

2023 **A. Guseva** and S. M. Tobias.
Transition to chaos and modal structure of magnetized Taylor–Couette flow
Philosophical Transactions of the Royal Society A, 381 (2023).
https://doi.org/10.1098/rsta.2022.0120

2022 **A. Guseva** and J. Jiménez.
Linear instability and resonance effects in large-scale opposition flow control
Journal of Fluid Mechanics, 935, A35 (2022).
https://doi.org/10.1017/jfm.2022.34

2020 J. I. Ibrahim, **A. Guseva**, and R. Garcia-Mayoral.
Selective opposition-like control of large-scale structures in wall-bounded turbulence
Journal of Physics: Conference Series. Vol. 1522. No. 1. (2020)
https://doi.org/10.1088/1742-6596/1522/1/012015

F. Nauman and **A. Guseva**.
Energy transfers in magnetohydrodynamic shear turbulence
Journal of Physics: Conference Series. Vol. 1522. No. 1. (2020)
http://doi.org/10.1088/1742-6596/1522/1/012005

2018 G. Mamatsashvili, F. Stefani, **A. Guseva**, M. Avila
Nonlinear evolution of helical magnetorotational instability in a magnetized Taylor-Couette flow
New Journal of Physics 20, 013012 (2018)
https://doi.org/10.1088/1367-2630/aa9d65

2017 **A. Guseva**, A.P. Willis, R. Hollerbach, M. Avila
Dynamo action in a quasi-Keplerian Taylor-Couette flow
Physical Review Letters, 119, 164501
https://doi.org/10.1103/PhysRevLett.119.164501

A. Guseva, A.P. Willis, R. Hollerbach, M. Avila
Transport properties of the azimuthal magnetorotational instability
The Astrophysical Journal, 849:92 (2017)
https://doi.org/10.3847/1538-4357/aa917d

A. Guseva, A.P. Willis, R. Hollerbach, M. Avila
Azimuthal magnetorotational instability at low and high magnetic Prandtl numbers
Magnetohydrodynamics, 53, 1:25-34 (2017).
https://arxiv.org/abs/1611.07296

2016 M. Gellert, G. Rüdiger, M. Schultz, **A. Guseva**, R. Hollerbach
Nonaxisymmetric MHD instabilities of Chandrasekhar states in Taylor-Couette geometry
Astrophysical Journal, 823, 99:1-9 (2016).
https://doi.org/10.3847/0004-637X/823/2/99

2015 **A. Guseva**, A.P. Willis, R. Hollerbach, M. Avila
Transition to magnetorotational turbulence in Taylor-Couette flow with imposed azimuthal magnetic field
New Journal of Physics, 17, 093018:1-14 (2015).
https://doi.org/10.1088/1367-2630/17/9/093018

A. Kharicha, I. Teplyakov, Yu. Ivochkin, M. Wu, A. Ludwig, **A. Guseva**
Experimental and numerical analysis of free surface deformation in an electrically driven flow
Experimental Thermal and Fluid Science, 62, 192-201 (2015).
http://www.sciencedirect.com/science/article/pii/S0894177714002933

Y. Ivochkin, I. Teplyakov, **A. Guseva**, D. Vinogradov
Influence of the swirled electrovortex flow on the melting of eutectic alloy In-Ga-Sn
Magnetohydrodynamics, 51, 2:337-344 (2015).
http://mhd.sal.lv/contents/2015/2/MG.51.2.18.R.html

Teaching experience

2026-ongoing Physics: oscillations and waves (*60h, undergraduate*)
Lectures for second-year undergraduate students in engineering
Barcelona School of Civil Engineering, UPC

2026-ongoing Thermodynamics (*60h, undergraduate*)
Lectures for second-year undergraduate students in aerospace engineering
Castelldefels School of Telecommunications and Aerospace Engineering, UPC

2025-2026 Thermodynamics and heat transfer (*180h, undergraduate*)
Lectures for second-year undergraduate students in engineering
Barcelona East School of Engineering, UPC, Spain

2025 **Dynamo Theory in Astrophysics** (*∼10h, PhD/researchers*)
Lectures on turbulence and mean-field dynamo theory
Institute of Astrophysics of Andalusia (IAA-CSIC), Granada, Spain
https://indico.iaa.csic.es/event/33/?print=1&view=lecture

2022-2025 **Magnetohydrodynamic instabilities and dynamo action** (*∼40h, master*)
Lectures and small research projects for master students in astrophysics as a part of the Gas Dynamics course
The Paris Observatory, France

2023 Numerical Analysis course at the School of Mathematics (*∼20h, undergraduate*)
Workshops for second-year undegraduate students in mathematics
University of Leeds, UK

2014-2015 Numerical methods for fluid mechanics (*∼40h, master*)
Seminars for graduate students in Mechanical, Chemical, Computational Engineering.
University of Erlangen-Nuremberg, Germany

Supervision of students

2026-ongoing — Júlia Muñoz
Undergraduate final year project "Interaction of fossil magnetic fields with surface convection in OBA stars" at UPC BarchelonaTech

2026 — Nicolas Barbier
Master-level research project "Reconstruction of stellar convective dynamos with convolutional neural networks" at UPC BarchelonaTech & ECAM, Belgium
`https://pmb.ecam.be/opac_css/index.php?lvl=author_see&id=5616`

2025 — Lamia Youbi
Undegraduate research project "Modelling of magnetic maps of protostars from DNS simulations" at IPAG, Université Grenoble Alpes
co-supervision with E. Alecian (30% involvement)

2025 — Maxime Bidoli
Master-level research project "Influence of accretion on convective dynamos in pre-main-sequence stars" at LIRA, the Paris Observatory
co-supervision with L. Petitdemange (50% involvement)

2024-2025 — Elsa Louaas
Master-level research project "Modelling of convective-radiative interfaces using transpiring boundary condtions" at LERMA, the Paris Observatory

2023-ongoing — Virgin Durepaire
PhD project "Magnetic instabilities in radiative stellar interiors"
at LIRA, the Paris Observatory
co-supervision with L. Petitdemange and K. Belkacem (30% involvement)

2023 — Joseph Hall
Undergraduate project "DMD for identification of topological waves in plasmas", in collaboration between U. Leeds and Brown University

2019-2020 — Joseph Ibrahim
Expert supervision of the project "Selective opposition-like control of large-scale structures in wall-bounded turbulence", in collaboration between Polytechnic University of Madrid and University of Cambridge (PhD with Dr. R. Garcia-Mayoral)

2014-2015 — Mohammed Ali Safari Shalmani
Master-level project "Parameter bounds of magnetorotational instability",
at University of Erlangen-Nuremberg

Selected talks and presentations

>30 first-author conference and seminar presentations, of which 11 invited talks

08/2026 — A. Guseva, L. Manchon, L. Petidemange, and C. Pinçon
Radial differential rotation leading to dipole collapse in pre-main-sequence stars
PLATO Stellar Science Conference, Liege, Belgium

06/2026 — Plenary talk (*40 selected presentations from nearly 400*)
Radial differential rotation leading to dipole collapse in pre-main-sequence stars
International conference "Cool stars 23", Tokyo, Japan

07/2026 — A. Guseva, L. Manchon, L. Petidemange, and C. Pinçon
Radial differential rotation leading to dipole collapse in pre-main-sequence stars
XVII Scientific Meeting of the Spanish Astronomical Society, Tarragona, Spain

07/2026 A. Guseva, C. Skene and S. Tobias
Data-driven discovery of dynamo equations
Workshop "Perspectives and challenges in Geophysical Fluid Dynamics", ENS, Paris

06/2026 A. Guseva, C. Skene and S. Tobias
Data-driven discovery of dynamo equations
Programme "Idealised mathematical models for geophysical flows", Institut Henri Poincare, Paris, France

04/2026 Invited talk
Data-driven discovery of dynamo equations
Seminars of the program Consequences Of Fields and Flows in the Interior and Exterior of the Sun (COFFIES - NASA-funded Heliophysics Center, USA)

11/2025 Invited talk
Magnetic fields in protostars: generation mechanisms and stability
Seminars of the program Consequences Of Fields and Flows in the Interior and Exterior of the Sun (COFFIES - NASA-funded Heliophysics Center, USA)

09/2025 Invited talk
Astrophysical magnetohydrodynamic flows: data-driven perspective
Seminars of the Laboratoire J.A. Dieudonné, UniCA, Nice, France

01/2025 Invited talk
Magnetic field dynamics in low mass stars
Severo Ochoa Colloquium, Institute of Astrophysics of Andalusia (IAA-CSIC), Granada, Spain

12/2024 Invited talk
Magnetic field dynamics in low mass stars
Seminars of the Institut d'Astrophysique Spatiale (IAS), Paris, France

09/2024 Invited talk
Large scales in turbulent shear flows
EEBE Seminars on Computational Mechanics, Barcelona, Spain

04/2024 Invited talk
Data-driven approach to interaction between magnetic fields and rotating convection
FAST-LISN Seminar of mechanics, Paris, France

03/2024 A. Guseva, C. Skene, S. Tobias
"Reduced-order modelling of dynamo waves: Sparse identification vs. weakly nonlinear analysis"
1st ERCOFTAC Workshop on Machine Learning for Fluid Dynamics, Sorbonne, Paris

09/2023 A. Guseva, Ludovic Petitdemange, S. Tobias
Weak and strong dynamos: a data-driven analysis
Geophysical & Astrophysical Fluid Dynamics: Experiments and Models Nice, France

07/2023 A. Guseva, S. Tobias
Data-driven analysis of magnetorotational turbulence
National Astronomy Meeting, Cardiff, UK

11/2022 Invited talk
A data-driven approach to nonlinear dynamos
Seminars of the Institute of Space Sciences, Barcelona, Spain

10/2022 A. Guseva, S. Tobias
Data-driven reduced-order modelling of dynamo waves
Programme "Frontiers in dynamo theory: from the Earth to the stars", Isaac Newtown Insititute, Cambridge, UK

Reviewer duties

I served as a reviewer for the German Research Foundation (DFG), Dutch Research Council (NWO), UK Engineering and Physical Sciences Research Council (EPSRC), and for several high-quality peer-reviewed journals:

- Monthly Notices of the Royal Astronomical Society
- Journal of Plasma Physics
- Journal of Fluid Mechanics
- Physical Review E
- Physical Review Fluids
- European Journal of Fluid Mechanics
- International Journal of Heat and Fluid Flow

Organization duties

2026 Initiative and organization of **Barcelona Astrophysical Fluid Dynamics Workshop**, bringing together local and international astrophysical fluids community 12-13 of May 2026, Barcelona, UPC
`https://shorturl.at/y4PnL`

2025 Representative for postdoctoral researchers at LIRA Laboratory Council, the Paris Observatory
Objectives: increasing postdoc visibility, facilitating interaction with the direction, and supporting career progression

2025 Member of the scientific committee of the 3rd Spanish Fluid Mechanics Conference (SFMC25)
`https://sfmc25.uma.es/organizing-committee/`

2025 Chair of the session "Waves and Coherent Structures" at the 23rd International Couette-Taylor Workshop (ICTW 2025), Durham, UK
`https://maths.dur.ac.uk/ICTW2025/programme.html`

2023 Organization of the interdisciplinary session "Magnetised stars and planets: combining theory and observations", National Astronomy Meeting, Cardiff, UK

2021-2023 Organization of seminars of Astrophysical and Geophysical Fluid Dynamics group, School of Mathematics, University of Leeds

2019 Organization of the 4th Summer School on Turbulence, Madrid, Spain